

Problem 1: The Geometry of Momentum

In Homework 1, you analyzed the eigenvalues of the momentum update matrix. Now, we will apply that intuition to a specific 1-dimensional "toy" problem.

We are trying to minimize $\mathcal{L}(w) = \frac{1}{2}\sigma^2 w^2$ using Gradient Descent with Momentum. The update rule is:

$$\begin{aligned}w_{t+1} &= w_t - \eta z_{t+1} \\z_{t+1} &= (1 - \beta)z_t + \beta \nabla \mathcal{L}(w_t)\end{aligned}$$

(a) **[System Dynamics]** We can write the state evolution as a linear system:

$$\begin{bmatrix} z_{t+1} \\ w_{t+1} \end{bmatrix} = R \begin{bmatrix} z_t \\ w_t \end{bmatrix}$$

Fill in the four entries of the matrix R in terms of β, η , and σ .

Solution:

First, compute the gradient: $\nabla \mathcal{L}(w_t) = \sigma^2 w_t$. Substitute this into the update for z :

$$z_{t+1} = (1 - \beta)z_t + \beta \sigma^2 w_t$$

Now substitute z_{t+1} into the update for w :

$$\begin{aligned}w_{t+1} &= w_t - \eta[(1 - \beta)z_t + \beta \sigma^2 w_t] \\w_{t+1} &= -\eta(1 - \beta)z_t + (1 - \eta\beta\sigma^2)w_t\end{aligned}$$

Putting this into matrix form:

$$R = \begin{bmatrix} (1 - \beta) & \beta\sigma^2 \\ -\eta(1 - \beta) & 1 - \eta\beta\sigma^2 \end{bmatrix}$$

Rubric: 5 pts total. 2 pts for correct z row. 3 pts for correct w row (must show substitution).

Problem 2: The "Lazy" Regularizer

Your teammate wants to train a linear model $Xw \approx y$ with **Weight Decay**, but the specialized hardware accelerator you are using only supports standard Stochastic Gradient Descent (SGD) on the Mean Squared Error (MSE) loss, with no option to add a separate L_2 penalty term to the loss function.

The available hardware API is: `HardwareSGD(input_batch, target_batch, learning_rate)`

You recall from Homework 1 that you can mimic regularization by injecting noise.

(a) **[Input Noise]** You propose adding random Gaussian noise $\xi \sim \mathcal{N}(0, \nu^2 I)$ to the **input** x during training. If you want to achieve an effective Weight Decay equivalent to minimizing:

$$\mathcal{L}_{reg}(w) = \frac{1}{m} \|Xw - y\|^2 + \lambda \|w\|^2$$

What variance ν^2 should you choose for your noise? Express ν^2 in terms of λ and/or the dataset size m .

Solution:

We analyze the expected loss with noisy input $x' = x + \xi$:

$$E[\mathcal{L}] = E \left[\sum_{i=1}^m ((x_i + \xi_i)^T w - y_i)^2 \right]$$

Expanding the quadratic:

$$= \sum_{i=1}^m (x_i^T w - y_i)^2 + \sum_{i=1}^m E[(\xi_i^T w)^2] + \text{Cross Terms (0)}$$

The expected noise term is:

$$E[w^T \xi_i \xi_i^T w] = w^T E[\xi_i \xi_i^T] w = w^T (\nu^2 I) w = \nu^2 \|w\|^2$$

So total expected loss is:

$$\|Xw - y\|^2 + m\nu^2 \|w\|^2$$

To match the target objective $\frac{1}{m} \|Xw - y\|^2 + \lambda \|w\|^2$, we divide our derived loss by m :

$$\frac{1}{m} \|Xw - y\|^2 + \nu^2 \|w\|^2$$

Matching coefficients, we require $\nu^2 = \lambda$. **Rubric:** 5 pts total. 2 pts for expanding the expected loss. 3 pts for the final relation $\nu^2 = \lambda$.

(b) **[Label Noise]** Your teammate suggests that instead of adding noise to the input, it might be computationally cheaper to add noise to the **labels** y . They propose: $\tilde{y} = y + \epsilon$, where $\epsilon \sim \mathcal{N}(0, \gamma^2)$.

Does this achieve L_2 regularization (Weight Decay) on w ? If yes, what is the effective λ ? If no, mathematically describe what effect this noise has on the expected loss function.

Solution:

No, it does not achieve regularization. Expected loss:

$$\begin{aligned} E[||Xw - (y + \epsilon)||^2] &= E[||(Xw - y) - \epsilon||^2] \\ &= ||Xw - y||^2 - 2(Xw - y)^T E[\epsilon] + E[||\epsilon||^2] \end{aligned}$$

Since $E[\epsilon] = 0$, the middle term vanishes. The last term is a constant (variance of noise) independent of w .

$$= ||Xw - y||^2 + \text{Constant}$$

Minimizing this is equivalent to minimizing the original un-regularized loss.

Rubric: 4 pts

total. 2 pts for "No". 2 pts for showing it only adds a constant to the loss.

Problem 3: SGD Memory

We often assume SGD converges to the minimum norm solution in the "interpolation regime" (where X is wide and we can fit data perfectly). However, this depends strictly on initialization.

Consider a simple 2D problem with 1 data point:

$$X = \begin{bmatrix} 1, & 2 \end{bmatrix} \quad y = [5]$$

We want to solve $Xw = y$, i.e., $w_1 + 2w_2 = 5$.

(a) **[The Standard Case]** If we initialize $w_{init} = [0, 0]^\top$, calculate the final solution w_∞ that SGD will converge to.

Solution:

When initialized at 0, SGD converges to the minimum L_2 norm solution. We want to minimize $w_1^2 + w_2^2$ subject to $w_1 + 2w_2 = 5$. Using Lagrange multipliers or geometry (projection of 0 onto the line): The vector is normal to the line, so $w \propto [1, 2]^\top$. $w = c[1, 2]^\top = [c, 2c]^\top$. Substitute into constraint: $c + 2(2c) = 5 \implies 5c = 5 \implies c = 1$. $w_\infty = [1, 2]^\top$. **Rubric:** 3 pts. 1 pt for recognizing min-norm. 2 pts for correct vector $[1, 2]$.

(b) **[The "Bad" Initialization]** Suppose we initialize $w_{init} = [5, 0]^\top$. Since $w_{init} = [5, 0]^\top$ is already a solution (check: $1(5) + 2(0) = 5$), SGD will not move. Is this the **minimum norm** solution? If not, calculate the squared Euclidean distance between this solution and the true minimum norm solution from part (a).

Solution:

No, it is not the min-norm solution (norm is $\sqrt{25} = 5$, whereas min-norm is $\sqrt{1+4} = \sqrt{5}$). Squared Distance:

$$\begin{aligned} \|[5, 0]^\top - [1, 2]^\top\|^2 &= \|[4, -2]^\top\|^2 \\ &= 4^2 + (-2)^2 = 16 + 4 = 20 \end{aligned}$$

Rubric: 3 pts. 1 pt for "No". 2 pts for correct distance calculation.

(c) **[The General Case]** Suppose we initialize at a non-solution point $w_{init} = [4, 0]^\top$. SGD updates always add a vector proportional to the row(s) of X . Using the fact that $w_\infty - w_{init} = \alpha X^\top$ for some scalar α , calculate the final vector w_∞ .

Solution:

The SGD update is always in the span of the rows of X . So the final solution is the orthogonal projection of w_{init} onto the solution line. Ansatz: $w_\infty = w_{init} + \alpha X^\top = [4, 0]^\top + \alpha[1, 2]^\top = [4 + \alpha, 2\alpha]^\top$. We know w_∞ must satisfy the constraint $Xw_\infty = 5$:

$$1(4 + \alpha) + 2(2\alpha) = 5$$

$$4 + 5\alpha = 5 \implies 5\alpha = 1 \implies \alpha = 0.2$$

Substitute α back:

$$w_\infty = [4 + 0.2, 2(0.2)]^\top = [4.2, 0.4]^\top$$

Rubric: 4 pts. 2 pts for setting up the projection/ansatz. 2 pts for final vector.

Problem 4: Bayesian Twists

In Homework 1, you proved that Ridge Regression is equivalent to MAP estimation with a Gaussian Prior $w \sim \mathcal{N}(0, \tau^2 I)$ and Gaussian Noise $n \sim \mathcal{N}(0, \sigma^2 I)$.

We are going to modify the probabilistic assumptions. For each modification below, identifying the new corresponding Loss Function terms.

Structure your answer in the form:

$$\operatorname{argmin}_w \quad ||Xw - y||^2 + \dots$$

(a) **[Biased Prior]** We believe the weights should be close to a specific "guess" vector w_0 , rather than 0.

$$\text{Prior: } w \sim \mathcal{N}(w_0, \tau^2 I)$$

What is the new Regularization term?

Solution:

The prior term in negative log likelihood is $-\log P(w)$.

$$P(w) \propto \exp\left(-\frac{||w - w_0||^2}{2\tau^2}\right)$$

Taking $-\log$:

$$\dots + \lambda ||w - w_0||^2$$

(Where λ depends on variance ratios, but we just need the form). **Rubric:** 3 pts. Correct term $||w - w_0||^2$.

(b) **[Heteroscedastic Noise]** We trust some data points more than others. The noise for the i -th data point has its own variance σ_i^2 .

$$y_i = w^\top x_i + n_i, \quad n_i \sim \mathcal{N}(0, \sigma_i^2)$$

How does the **Reconstruction Loss** term ($||Xw - y||^2$) change? Write the new sum over $i = 1 \dots m$.

Solution:

The likelihood term $-\log P(y|X, w)$ changes.

$$P(y_i|x_i, w) \propto \exp\left(-\frac{(y_i - w^\top x_i)^2}{2\sigma_i^2}\right)$$

Summing over all independent data points:

$$\operatorname{argmin}_w \sum_{i=1}^m \frac{1}{\sigma_i^2} (y_i - w^\top x_i)^2 + \dots$$

This is Weighted Least Squares. **Rubric:** 3 pts. Correct weighted sum formulation.